

# Selenium WebDriver UI Automation Guide

Comprehensive Java Reference Implementation for Standard Web Interactions

## SCENARIO 1

### Select Favorite IPL Team from Dropdown Menu

Static HTML `<select>` elements are best handled using Selenium's dedicated `Select` wrapper class, which provides specialized methods to pick options by visible text, index, or value.

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.Select;

public class DropdownAutomation {
    public static void main(String[] args) {
        WebDriver driver = new ChromeDriver();
        try {
            driver.get("https://example.com/sports-fan-signup");

            // Locate the HTML <select> element
            WebElement teamDropdown = driver.findElement(By.id("ipl-team-select"));

            // Initialize the Select wrapper class
            Select selectTeam = new Select(teamDropdown);

            // Select option by visible text
            selectTeam.selectByVisibleText("Mumbai Indians");

            // Verification
            String selectedOption = selectTeam.getFirstSelectedOption().getText();
            System.out.println("Selected IPL Team: " + selectedOption);

        } finally {
            driver.quit();
        }
    }
}
```

## SCENARIO 2

### Check and Uncheck Multiple Cuisine Checkboxes

Before executing a click command on checkbox controls, always inspect the state via `isSelected()` to prevent toggling an already active checkbox.

```

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class CheckboxAutomation {
    public static void main(String[] args) {
        WebDriver driver = new ChromeDriver();
        try {
            driver.get("https://example.com/food-delivery-form");

            String[] targetCuisines = {"Pizza", "Biryani"};

            for (String cuisine : targetCuisines) {
                // Find checkbox by value or associated attribute
                WebElement checkbox = driver.findElement(
                    By.xpath("//input[@type='checkbox' and @value='" + cuisine + "']")
                );

                // Check the box if not already selected
                if (!checkbox.isSelected()) {
                    checkbox.click();
                    System.out.println(cuisine + " checked.");
                }

                // Uncheck the box
                if (checkbox.isSelected()) {
                    checkbox.click();
                    System.out.println(cuisine + " unchecked.");
                }
            }
        } finally {
            driver.quit();
        }
    }
}

```

### SCENARIO 3

## Select and Verify Payment Method Radio Buttons

Radio buttons in a payment checkout flow operate within mutually exclusive groups. Activating a option toggles off any previously active radio button.

```

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class RadioButtonAutomation {
    public static void main(String[] args) {
        WebDriver driver = new ChromeDriver();
        try {
            driver.get("https://example.com/checkout");

            // Locate the UPI radio button
            WebElement upiOption = driver.findElement(By.id("payment-upi"));

            // Click to select UPI
            upiOption.click();

            // Assertion / Verification step
            if (upiOption.isSelected()) {
                System.out.println("Verification Passed: UPI option is successfully selected.");
            } else {
                System.out.println("Verification Failed: UPI option is not selected.");
            }
        } finally {
            driver.quit();
        }
    }
}

```

#### SCENARIO 4

### Pick a Specific Date from a Date Picker

Standard HTML5 date pickers can accept input formatted as strings via `sendKeys()`. Custom calendar widgets require interacting directly with target DOM day cells.

```

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class DatePickerAutomation {
    public static void main(String[] args) {
        WebDriver driver = new ChromeDriver();
        try {
            driver.get("https://example.com/movie-booking");

            // Direct input approach (for <input type="date">)
            WebElement dateInput = driver.findElement(By.id("dob-picker"));
            dateInput.clear();
            dateInput.sendKeys("15-08-1998"); // Format dependent on UI locale

            /* Alternative Calendar Overlay Click Approach:
            dateInput.click();
            driver.findElement(By.xpath("//td[@data-date='15']")).click();
            */

            System.out.println("Date selected successfully: " + dateInput.getAttribute("value"));
        } finally {
            driver.quit();
        }
    }
}

```

#### SCENARIO 5

### Auto-suggest Input Traversal & Verification

Dynamic input fields with auto-suggestions require explicit synchronization (`WebDriverWait`) to handle asynchronous element rendering before keyboard or element clicks are executed.

```

import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class AutoSuggestAutomation {
    public static void main(String[] args) {
        WebDriver driver = new ChromeDriver();
        try {
            driver.get("https://example.com/zomato-search");

            WebElement searchBox = driver.findElement(By.xpath("//input[@placeholder='Search city']"));
;
            searchBox.sendKeys("Bung");

            // Explicit wait for dynamic suggestion items to appear in DOM
            WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(5));
            wait.until(ExpectedConditions.visibilityOfElementLocated(
                By.cssSelector("ul.suggestions-list > li")
            ));

            // Method A: Keyboard arrow navigation to select 2nd suggestion
            searchBox.sendKeys(Keys.ARROW_DOWN);
            searchBox.sendKeys(Keys.ARROW_DOWN);
            searchBox.sendKeys(Keys.ENTER);

            /* Method B: Direct locator approach (Alternative)
            List<WebElement> suggestions = driver.findElements(By.cssSelector("ul.suggestions-list >
li"));
            suggestions.get(1).click(); // Index 1 represents the 2nd suggestion
            */

            // Verification step
            String enteredValue = searchBox.getAttribute("value");
            System.out.println("Verified entered city: " + enteredValue);

        } finally {
            driver.quit();
        }
    }
}

```